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=====
MiniRTOS
ARM Cortex-M4 RTOS
=====
```

System Initialization...

```
[OK] UART Driver
[OK] SysTick Timer
[OK] Scheduler
[OK] Kernel Memory
[OK] Task IPC Objects
```

```
=====
MiniRTOS Boot
=====
```

```
=====
Tasks Created Successfully
Starting Scheduler...
=====
```

```
[LED] Task Running
[UART] Task Running
[SENSOR] Task Running
[SEM] Sender Giving Signal
[MUTEX] Task1 Locked
[QUEUE] Sent: 0
[QUEUE] Received: 0
[SEM] Receiver Received Signal
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	PRIORITY	STATE
APP	LOW	RUNNING
MEMORY	LOW	READY
SEM_RX	MED	BLOCKED
QUEUE_RX	MED	BLOCKED
QUEUE_TX	MED	BLOCKED
MUTEX2	MED	BLOCKED
MUTEX1	MED	BLOCKED
SEM_TX	MED	BLOCKED
SENSOR	MED	BLOCKED
UART	MED	BLOCKED
LED	HIGH	BLOCKED

```
-----
[MEMORY] Task Running
[MUTEX] Task1 Released
[MUTEX] Task2 Waiting
[SEM] Receiver Received Signal
[LED] Task Running
[SENSOR] Task Running
[UART] Task Running
[SEM] Sender Giving Signal
[QUEUE] Sent: 1
[SEM] Receiver Received Signal
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	PRIORITY	STATE
APP	LOW	RUNNING
SEM_RX	MED	BLOCKED
QUEUE_RX	MED	BLOCKED
QUEUE_TX	MED	BLOCKED
SEM_TX	MED	BLOCKED
UART	MED	BLOCKED
SENSOR	MED	BLOCKED
LED	HIGH	BLOCKED
MUTEX2	MED	BLOCKED
MUTEX1	MED	BLOCKED
MEMORY	LOW	BLOCKED

```
=====
[MUTEX] Task2 Acquired
[QUEUE] Received: 1
[LED] Task Running
[UART] Task Running
[SENSOR] Task Running
[MUTEX] Task2 Released
[SEM] Sender Giving Signal
[QUEUE] Sent: 2
[SEM] Receiver Received Signal
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	PRIORITY	STATE
APP	LOW	RUNNING
MEMORY	LOW	READY
QUEUE_RX	MED	BLOCKED
SEM_RX	MED	BLOCKED
QUEUE_TX	MED	BLOCKED
MUTEX1	MED	BLOCKED
SEM_TX	MED	BLOCKED
MUTEX2	MED	BLOCKED
SENSOR	MED	BLOCKED
UART	MED	BLOCKED
LED	HIGH	BLOCKED

```
-----
[MEMORY] Task Running
[QUEUE] Received: 2
[SEM] Receiver Received Signal
[LED] Task Running
[SENSOR] Task Running
[UART] Task Running
[MUTEX] Task1 Locked
[SEM] Sender Giving Signal
[QUEUE] Sent: 3
[SEM] Receiver Received Signal
[QUEUE] Received: 3
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	PRIORITY	STATE
APP	LOW	RUNNING
QUEUE_RX	MED	BLOCKED
SEM_RX	MED	BLOCKED
QUEUE_TX	MED	BLOCKED
MUTEX2	MED	BLOCKED
SEM_TX	MED	BLOCKED
MUTEX1	MED	BLOCKED
UART	MED	BLOCKED
SENSOR	MED	BLOCKED
LED	HIGH	BLOCKED
MEMORY	LOW	BLOCKED

```
=====
[MUTEX] Task1 Released
[MUTEX] Task2 Waiting
[LED] Task Running
[UART] Task Running
[SENSOR] Task Running
[SEM] Sender Giving Signal
[SEM] Receiver Received Signal
[QUEUE] Sent: 4
[MEMORY] Task Running
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	PRIORITY	STATE
APP	LOW	RUNNING
MEMORY	LOW	BLOCKED
QUEUE_RX	MED	BLOCKED
QUEUE_TX	MED	BLOCKED
SEM_RX	MED	BLOCKED
SEM_TX	MED	BLOCKED
SENSOR	MED	BLOCKED
UART	MED	BLOCKED
LED	HIGH	BLOCKED
MUTEX2	MED	BLOCKED
MUTEX1	MED	BLOCKED

```
-----
[SEM] Receiver Received Signal
[MUTEX] Task2 Acquired
[QUEUE] Received: 4
[LED] Task Running
[SENSOR] Task Running
[UART] Task Running
[MUTEX] Task2 Released
[SEM] Sender Giving Signal
[SEM] Receiver Received Signal
[QUEUE] Sent: 5
[QUEUE] Received: 5
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	PRIORITY	STATE
APP	LOW	RUNNING
QUEUE_RX	MED	BLOCKED
QUEUE_TX	MED	BLOCKED
SEM_RX	MED	BLOCKED
MUTEX1	MED	BLOCKED
SEM_TX	MED	BLOCKED
MUTEX2	MED	BLOCKED
UART	MED	BLOCKED
SENSOR	MED	BLOCKED
LED	HIGH	BLOCKED
MEMORY	LOW	BLOCKED

```
=====
[SEM] Receiver Received Signal
[LED] Task Running
[UART] Task Running
[SENSOR] Task Running
[MUTEX] Task1 Locked
[SEM] Sender Giving Signal
[SEM] Receiver Received Signal
[QUEUE] Sent: 6
[MEMORY] Task Running
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	PRIORITY	STATE
APP	LOW	RUNNING
MEMORY	LOW	BLOCKED
QUEUE_RX	MED	BLOCKED
QUEUE_TX	MED	BLOCKED
SEM_RX	MED	BLOCKED
MUTEX2	MED	BLOCKED
SEM_TX	MED	BLOCKED
MUTEX1	MED	BLOCKED
SENSOR	MED	BLOCKED
UART	MED	BLOCKED
LED	HIGH	BLOCKED

```
=====
[SEM] Receiver Received Signal
[MUTEX] Task1 Released
[MUTEX] Task2 Waiting
[QUEUE] Received: 6
[LED] Task Running
[SENSOR] Task Running
[UART] Task Running
[SEM] Sender Giving Signal
[SEM] Receiver Received Signal
[QUEUE] Sent: 7
[QUEUE] Received: 7
[APP] Task Running
```

```
=====
MiniRTOS Task Monitor
=====
```

NAME	PRIORITY	STATE
APP	LOW	RUNNING
QUEUE_RX	MED	BLOCKED
QUEUE_TX	MED	BLOCKED
SEM_RX	MED	BLOCKED
SEM_TX	MED	BLOCKED
UART	MED	BLOCKED
SENSOR	MED	BLOCKED
LED	HIGH	BLOCKED
MUTEX2	MED	BLOCKED
MUTEX1	MED	BLOCKED
MEMORY	LOW	BLOCKED

```
-----
[SEM] Receiver Received Signal
```

```
[MUTEX] Task2 Acquired
```

```
[LED] Task Running
```

```
[UART] Task Running
```

```
[SENSOR] Task Running
```

```
[SEM] Sender Giving Signal
```

```
[SEM] Receiver Received Signal
```

```
[MUTEX] Task2 Released
```

```
[QUEUE] Sent: 8
```

```
[MEMORY] Task Running
```

```
[APP] Task Running
```